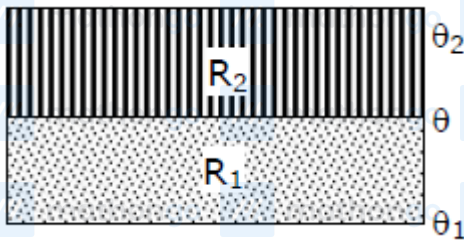


Q1: 25 Feb (Shift 2) - Numerical

A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52 K, its efficiency is doubled. The temperature in Kelvin of the source will be

Q2: 26 Feb (Shift 1) - Single Correct

The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by :



(1) $\frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$

(2) $\frac{\theta_1 R_2 - \theta_2 R_1}{R_2 - R_1}$

(3) $\frac{\theta_2 R_2 - \theta_1 R_1}{R_2 - R_1}$

(4) $\frac{\theta_1 R_1 + \theta_2 R_2}{R_1 + R_2}$

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Answer Key

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Q1 (208)

Q2 (1)

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Q1 (208)

$$\therefore \eta = \frac{W}{Q_{\text{in}}} = \frac{1}{4}$$

$$\frac{1}{4} = 1 - \frac{T_1}{T_2}$$

$$\frac{T_1}{T_2} = \frac{3}{4}$$

When the temperature of the sink is reduced by 52k then its efficiency is doubled, $\frac{1}{2} = 1 - \frac{(T_1 - 52)}{T_2}$

$$\frac{T_1 - 52}{T_2} = \frac{1}{2}$$

$$\frac{T_1 - 52}{T_2} = \frac{1}{2}$$

$$\frac{3}{4} - \frac{52}{T_2} = \frac{1}{2}$$

$$\frac{52}{T_2} = \frac{1}{4}$$

$$T_2 = 208\text{k}$$

Q2 (1)

Temperature at the junction is θ .

so using the formula

$$\frac{T_2 - T}{R_1} = \frac{T - T_1}{R_2}$$

$$\frac{\theta_2 - \theta}{R_2} = \frac{\theta - \theta_1}{R_1}$$

$$R_1(\theta_2 - \theta) = R_2(\theta - \theta_1)$$

$$R_1\theta_2 - R_1\theta = R_2\theta - R_2\theta_1$$

$$R_1\theta + R_2\theta = R_1\theta_2 + R_2\theta_1$$

$$\theta = \frac{R_1\theta_2 + R_2\theta_1}{R_1 + R_2}$$